IDRIS SADIQ

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PROFILE & RESEARCH INTERESTS

Electrical and Electronics Engineering undergraduate with hands-on experience in embedded hardware, sensing, and control, and a strength in synthesizing ideas across domains within a team. Drawn to compact robotic systems; preparing to apply for a Master's in Japan (October 2027).

EDUCATION

SRM Easwari Engineering College, Chennai, India Aug 2023 – May 2027 (Expected)
B.E. in Electrical and Electronics Engineering (Minor in Robotics) - CGPA: 8.42 / 10

LANGUAGES

Japanese: JLPT N5 & NAT Q4 certified; currently working toward JLPT N3. English: Fluent.

RESEARCH & TECHNICAL EXPERIENCE

Research Intern - ABB Global Centre of Excellence in Robotics, SRM Easwari Engineering College Jul 2026 – Present

- Working on an ABB IRB 1600 in a 3D bin-picking cell (SICK vision + PLB engine), validated first in ABB RobotStudio.
- Contributing as a cross-domain integrator on an early-stage RL-based PID auto-tuning project for an industrial robot under dynamic payload variation.

Electronics and Robotics Intern - Subtlebotic, Chennai Sep 2025 – Nov 2025

- Drafted schematics and PCBs for boost converters, BMS and switch circuits; optimized BOM cost across 50+ parts.

RESEARCH & TECHNICAL PROJECTS

Final Year Project - Self-Balancing Swarm Robotic System with Adaptive Cooperative Formation Jul 2026 – Jan 2027

- Designing a decentralized swarm of three self-balancing robots coordinating via infrared and ESP-NOW (no camera, no GPS), with on-demand physical docking.
- Incorporating sun-sensor-based heading determination for GPS-free navigation, modelled on NASA's Mars rovers, for lunar/Martian deployment.

Gesture-Sensing IMU Wristband - "G.A.I.T. Assist" (EPICS in IEEE Grant, $3,000) Ongoing

- Designing and building the full hardware platform (sensor selection through embedded firmware) for a wearable IMU-based gesture-recognition wristband assisting Parkinson's patients with daily tasks and monitoring.
- Leading hardware development on a 4-person interdisciplinary team, with teammates handling gesture-classification software.

Quadruped Robots - Spider-Inspired and Minimal-Leg Designs Personal projects

- Built two quadrupeds: a spider-inspired leg design for moving over rubble, and an open-source minimal-leg design that walks by knee-lift and leg shifting.

stepscribe-cad - Open-Source CAD Analysis Tool (PyPI, AGPL-3.0) Oct 2026

- Wrote and published a Python package that reads CAD STEP files and produces plain-text descriptions of exact measurements, features, and assembly facts, so AI tools can reason about mechanical designs.

AI-Powered Beach Sand Mapping Rover - Smart India Hackathon Sep 2025

- Built a low-cost, computer-vision-based rover automating beach sand grain-size mapping across intertidal zones.

More projects (PID robot, custom PCB/BMS builds, 8-bit breadboard computer) on LinkedIn.

ACHIEVEMENTS & PUBLICATIONS

- First-author conference paper, presented at ICCORSE 2026: "Adaptive Swarm-Actuated Net System for Space Debris Capture: Integrating Distributed Robotics with Flexible Capture Mechanisms" (ASANS).
- National Finalist, India Innovates 2026 - ~20,000 participants; Bharat Mandapam, New Delhi (Mar 2026).
- Runner-up, InnovateX Hackathon - on-the-spot problem-solving round, 100+ teams, RIT & IEEE SPAx Regional.
- Co-authored & presented a conference abstract, ICUMSAT 2025: "Design and Optimization of a Rare-Earth-Free PM-Assisted Synchronous Reluctance Motor using Tetrataenite for Light Electric Vehicles".

SKILLS & CERTIFICATIONS

- Software: Python, C, SQL, Linux, Git, basic Computer Vision / AI; MATLAB, Fusion 360, ABB RobotStudio
- Hardware: PCB Design (Altium, EasyEDA), BMS, Embedded Systems, PID / LQR Control, IMU / sensor integration
- Certifications: NPTEL (Advanced/Industrial/Experimental Robotics, Computer Vision), Altium Designer Essentials